

	Document Title	Rev.	Project FFEx MoC	
	TECHNICAL SPECIFICATION CONSTRUCTION AND CONFIGURATION OF CCTV SYSTEM	0		

Novo Nordisk Document Number	Related Document Number
MOC4-08-9600-RFQ-004-AFRY - Annex 1	-

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Revision	Date	Des.	Check	Approv.	Emission Purpose
0	26/04/14	ugb	jap	lrm	IFC - Issued For Construction
1					
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9					
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NN Document Approver (Stamp or Signature)

General Notes

	Afry Document Number:	109004798-001-SITE-F-0414				Revision:	0
	Distribution:	Novo Nordisk - E	AFRY BR - E				

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1. INTRODUCTION

This document aims to establish the technical requirements and functional criteria for the construction, development and configuration of the entire CCTV camera system project, including the equipment installation, cabling, power supply, network connectivity, installation, licensing, updates, permissions, configuration, programming, alarm monitoring, display creation/configuration, third party integrations (Access Control System - Schneider Ecostruxure Security Expert), backups, testing (functional, acceptance, backup/retrieval, etc.) and operator training during the Executive Project Phase of the Expansion of the UB – Utility Building – C40, SP – Spine Building – S1, AP – Asseptic Production - AP1, Data centers (D20 and D21) and WH – Warehouse as well as External Areas, Entrances and Boiler at the industrial complex known as the MOC Site Expansion, owned by Novo Nordisk in Montes Claros - MG.

2. OBJECTIVE

This document aims to establish the basis for the Request for Quotation, with a view to contracting under a LUMP SUM contract:

- **CCTV Systems:** consists of all the installation services for cameras required for the expansion of the existing system, along with their respective licenses, to ensure real-time monitoring and management of images across the networks (CORP / PSNET), i.e., Security/Process, throughout the entire factory expansion, securely supporting the operation and communication of the CCTV.
- **CCTV Systems Configuration:** consists of all camera software installation and configuration, along with their respective licenses, to ensure real-time monitoring and management of images across the networks, throughout the entire factory expansion, securely supporting the operation and communication of the CCTV. The CCTV system will be **Avigilon Control Center**, in compliance with the standardization adopted by SITE MOC's Asset Security.

The project scope consists of:

- **Design and Engineering:** development of the executive project according to technical standards and customer requirements;
- **Supplies:** procurement of all building materials and accessories necessary for the implementation of all systems;
- **Installations:** deployment, fastening, connection, termination, organization, and identification of all cameras, as well as assembly of devices and minimum adjust of infrastructure for cameras cable accommodation;
- **Commissioning and Start-up:** testing and validation of all images in the Control Center ensuring the adjustments and quality of the images;
- **Assisted Operation:** post-installation support for the initial functioning of the system;

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- **Technical Team:** mobilization of a specialized team, including installers, site coordinator, occupational safety technician, planner, and other professionals necessary for the perfect implementation of the entire structured cabling infrastructure.
- **Installation of the Video Management Software:** This step includes deploying and configuring the CCTV video management platform on the designated servers or workstations. It ensures the system is properly installed, operational, and ready to receive camera streams and user configurations. This installation must be done under supervision of NovoNordisk's IT department.
- **Implementation of Licensing and Firmware Updates:** In this phase, all necessary software licenses must be activated, and camera firmware or system firmware must be updated to the latest supported versions. This guarantees compliance, stability, and access to the newest security features.
- **User and permission structure configuration:** Creation and configuration of operator accounts, administrative roles, access groups, and permission levels, ensuring that system privileges align with organizational policies and operational requirements.
- **Implementation of Cameras in the Systems:** This involves adding each camera to the management system, configuring IP addresses, streaming parameters, and assigning them to corresponding locations or maps. It ensures all video devices are recognized and integrated. This also includes programming of any necessary logic for the proper functioning of the system.
- **Integration of Existing Cameras into the Systems:** This involves adding and configuring the existing cameras of the Data Centers (D20 and D21) into the new video monitoring system. This must include any necessary licensing and communication as well as adding the cameras to the system, configuring IP addresses, streaming parameters, and assigning them to corresponding locations or maps. Each Datacenter has 5 internal cameras, 6 external cameras and 3 access control doors which must be integrated according to the item "Third Party Systems Integration" on this list.
- **Alarm monitoring:** Setup of alarm conditions where necessary.
- **Implementation and Configuration of Operator Workstation Displays:** This step includes setting up monitoring screens, video walls, or operator consoles, with layouts, view cycles, alarms, and permissions adjusted according to operational needs.
- **Implementation of Camera Analytics when Necessary:** Advanced video analytics—such as motion detection, intrusion detection, LPR, or object classification—are configured for cameras that require these features. This enhances security capabilities and enables automated event detection.
- **Third party system integrations:** Configuration and validation of integrations with external systems such as Access Control System for event association, or client-provided applications. The Access Control integration must include the relevant licensing for receiving and reacting to triggers from the Access Control System (EcoStruxure Security Expert) for specific monitoring conditions. The BIDDER must decide, implement and configure the communication protocols used for third party communications.

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The number of cameras integrated into the system shall be the same as the total number of doors monitored by the access control system.

- **Implementation of Image Backup System:** A backup mechanism is deployed to ensure recorded footage is securely duplicated, whether to secondary storage, cloud services, or long-term archival systems. This guarantees redundancy and preserves evidence.
- **Testing:** General functional tests are performed to validate image quality, camera connectivity, system performance, and behavior under normal and stress conditions. The goal is to confirm that the system meets operational requirements.
- **Recording, Storage and Image Retrieval Tests:** These tests validate the complete video lifecycle: recording continuity, storage integrity, retention time compliance, and successful retrieval of footage for investigation. It ensures the system performs reliably in real-world scenarios.
- **Operator and administrator training:** Training sessions for system operators and administrators covering system operation, camera management, alarm handling, reporting, and maintenance procedures as well as any other system functionalities included in this scope.

NOTES:

- For process cameras, it should be considered that the devices must be on-site at least 30 days prior to installation. This lead time is necessary to perform the bench test — an activity carried out by Novo Nordisk to configure the equipment. This test can be conducted under the supervision of the contracted company, as the devices remain under their responsibility during this period;
- It should have a pre-configuration as well for all cameras. It's aimed at updating firmware and assurance of functionalities of the cameras;
- All cameras must be configured with the correct tags and IPs as provided by the CLIENT;
- For all cameras the post-installation should be monitored by the area staff to assure the correct image and angle is captured in the cameras.

3. DEFINITIONS

The following definitions are established:

- **OWNER:** NOVO NORDISK BRASIL
- **CLIENT:** NOVO NORDISK BRASIL
- **MANAGER:** Company designated by the OWNER as its representative for the management of the implementation.
- **THIRD PARTIES:** Companies hired by the OWNER for the design and execution of predefined modules of the project.
- **BIDDERS:** Companies participating in this bidding process that are offering services, materials, and/or equipment.

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- **CONTRACTOR:** The company that wins the bid, which will carry out the work and provide the services, materials, and/or equipment for this bidding process.

4. STANDARDS

The standards and resolutions listed below were used in the preparation of this document or contain applicable instructions and procedures. They must be used in their latest revision.

The list below describes the standards and resolutions; however, the BIDDER must not be limited only to the items presented below. These are:

- **ABNT - Associação Brasileira de Normas Técnicas**
 - *NBR IEC 60529* *Graus de proteção providos por Invólucros (Código IP);*
 - *NBR IEC 61000* *Compatibilidade Eletromagnética (EMC);*
 - *NBR 14565* *Procedimento Básico para Elaboração de Projetos de Cabeamento de Telecomunicações para Rede Interna Estruturada;*
 - *NBR 5410* *Instalações Elétricas de Baixa Tensão;*
 - *NBR 5419* *Proteção Contra Descargas Atmosféricas;*
 - *NBR 13571* *Sistemas de Aterramento de Equipamentos de Telecomunicações.*
- **ISPE – International Society for Pharmaceutical Engineering**
 - *GAMP 5* *Good Automated Manufacturing Practice – A Risk-Based Approach to Compliant GxP Computerized Systems*
- **ISO – International Organization for Standardization**
 - *ISO/IEC 11801* *Information Technology - Generic Cabling for Customer Premises — Part 1: General requirements;*
 - *ISO/IEC 14763-2* *Information Technology - Implementation and Operation of Customer Premises Cabling – Part 2: Planning and Installation;*
 - *ISO/IEC JTC 1/SC 29* *Coding of Moving Pictures and Audio;*
 - *ISO/IEC 27001* *Information Security Management Systems*
- **ECIA - Electronic Components Industry Association**
 - *EIA 310-D Cabinets, Racks, Panels and Associated Equipment.*
- **ONVIF (Open Network Video Interface Forum)**

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- ONVIF Profile S *Streaming de vídeo e controle de PTZ.*
- ONVIF Profile G *Gravação e armazenamento.*
- ONVIF Profile T *Vídeo H.265 e metadados.*

- **ANSI - American National Standards Institute / TIA - Telecommunications Industry Association**

- ANSI/TIA-568.0-E *Generic Telecommunications Cabling for Customer Premises;*
- ANSI/TIA-568.1-E *Commercial Building Telecommunications Cabling Standard;*
- ANSI/TIA-568.2-D *Balanced Twisted-Pair Telecommunications Cabling and Components Standard;*
- ANSI/TIA-568.3-E *Optical Fiber Cabling and Components Standard;*
- ANSI/TIA-569-E *Telecommunications Pathways and Spaces;*
- ANSI/TIA-606-C *Administration Standard for Telecommunications Infrastructure;*
- ANSI/TIA-607-D *Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises;*
- ANSI/TIA-1005-A *Telecommunications Infrastructure Standard for Industrial Premises;*
- ANSI/TIA-1179-B *Healthcare Facility Telecommunications Infrastructure Standard;*
- ANSI/TIA-862-C *Structured Cabling Infrastructure Standard for Intelligent Building Systems.*

- **ITU – International Telecommunication Union**

- H.264 / H.265 / H.266 *International standards for video coding and compression - Efficient video storage and transmission and Compatibility across various CCTV equipment and software platforms.*

- **IEC - International Electrotechnical Commission**

- IEC 62676 *Video surveillance systems for use in security applications*
- ISO/IEC 14496-8 *Coding of Audio-Visual Objects - Part 8: Carriage of ISO/IEC 14496 contents over IP networks*
- ISO/IEC 14496-10 *Coding of Audio-Visual Objects - Part 10: Advanced Video Coding*
- IEC/EN 50081-2 *Electromagnetic Compatibility. Generic Emission Standard. Industrial Environment*

- **ANATEL - Agência Nacional de Telecomunicações**

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- Resolução nº 700 *Regulamento sobre a Avaliação da Exposição Humana a Campos Elétricos, Magnéticos e Eletromagnéticos Associados à Operação de Estações Transmissoras de Radiocomunicação;*
- Resolução nº 715 *Regulamento de Avaliação da Conformidade e de Homologação de Produtos para Telecomunicações.*

- **TELEBRAS - Telecomunicações Brasileiras S.A.**

- Prática 240-600-703 *Especificações Gerais das Condições Ambientais Aplicáveis a Equipamentos, Dispositivos e Acessórios de Telecomunicações;*
- Prática 240-360-703 *Especificações Gerais das Cabinas Transportáveis. ("CONTAINERS") para abrigar Sistemas de Telecomunicações.*

The OWNER requires the BIDDER to fully comply with the regulatory standards – NRs of the Consolidation of Labor Laws, related to Occupational Health and Safety, in accordance with ordinance 3214, of 06/08/1978 and its updates, as well as full compliance with local health and safety legislation in force.

Legal requirements shall always prevail over those contained in this document, except when the requirements herein are more restrictive.

All the above-mentioned standards must be consulted in their latest revision and must be in force; otherwise, the standard that replaced it should be used.

5. REFERENCE DOCUMENTS

The following documents apply:

GENERAL

-	CRITERIA CONTROL – IT TELECOMMUNICATIONS
350.ES.5913.01	ES – CCTV SURVEILLANCE – ENGINEERING SPECIFICATION
F-00891561	FS – CCTV SURVEILLANCE – FUNCIONAL SPECIFICATION
F-00891566	HDS – CCTV SURVEILLANCE – HARDWARE DESIGN SPECIFICATION
MOC4-C40-08-9600-ES-009-AFRY	TECHNICAL SPECIFICATION - STRUCTURED CABLING (CORP/PSNET) - DATA, VOICE, VIDEO AND WI-FI
MOC-SITE-06-DD-04-000000-F-01160300	NETWORK ARCHITECTURE (CORP/PSNET) - TELECOMMUNICATIONS SYSTEMS - 1/2
MOC-SITE-06-DD-04-00000-F-01160302	NETWORK ARCHITECTURE (CORP/PSNET) - TELECOMMUNICATIONS SYSTEMS – 2/2

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MOC-SITE-06-DD-04-000000-F-01122438	IT NETWORK ARCHITECTURE (CORP / PSNET) – CCTV SYSTEM
Q0763783	GITO - Global IT Infrastructure Standards
Q105336	PHYSICAL SECURITY IN NOVO NORDISK
MOC4-08-9600-RFQ-001-AFRY	REQUEST FOR QUOTE FOR CCTF CONTROL SYSTEM
MOC4-08-9600-RFQ-003-AFRY	REQUEST FOR QUOTE FOR STRUCTURED CABLING (CORP/PSNET) DATA, VOICE, VIDEO AND WI-FI SYSTEM
Q124747	PSNET INFRASTRUCTURE AND SERVICES GOVERNANCE
Q187655	MANAGE INFORMATION SECURITY IN IT SOLUTIONS
MOC-SITE-08-06-050000-F-01215240	TYPICAL TELECOMMUNICATIONS DETAILS COVER
MOC4-25-0962-ES-001	ESPECIFICAÇÃO DE REQUISITOS FUNCIONAIS PARA O SISTEMA DE CCTV PATRIMONIAL
MOC4-C40-08-9606-ES-010-AFRY	TECHNICAL SPECIFICATION CCTV SYSTEMS (CORP/PSNET) PROCESS AND SURVEILLANCE CAMERAS”

- **UB - UTILITY BUILDINGS (C40)**

MOC-C40-08-01-050000-F-01202138	IT NETWORK PLAN - CCTV POINTS PLACEMENT - 1ST FLOOR
MOC-C40-08-01-050000-F-01202139	IT NETWORK PLAN - CCTV POINTS PLACEMENT - 2ND FLOOR
MOC-C40-08-01-050000-F-01202140	IT NETWORK PLAN - CCTV POINTS PLACEMENT - 3RD FLOOR
MOC-C40-08-01-050000-F-01202142	IT NETWORK PLAN – CCTV POINTS PLACEMENT- 4TH FLOOR
MOC-C40-08-01-050000-F-01268121	IT NETWORK PLAN – CCTV POINTS PLACEMENT- 5TH FLOOR
MOC-C40-08-01-050000-F-01268123	IT NETWORK PLAN – CCTV POINTS PLACEMENT- BOILER
MOC-C40-08-01-050000-F-01202167	IT NETWORK PLAN - CCTV ROUTING (CORP/PSNET) - 1ST FLOOR
MOC-C40-08-01-050000-F-01202169	IT NETWORK PLAN - CCTV ROUTING (CORP/PSNET) - 2ND FLOOR
MOC-C40-08-01-050000-F-01202170	IT NETWORK PLAN - CCTV ROUTING (CORP/PSNET) - 3RD FLOOR
MOC-C40-08-01-050000-F-01202171	IT NETWORK PLAN - CCTV ROUTING (CORP/PSNET) - 4TH FLOOR
MOC-C40-08-01-050000-F-01268130	IT NETWORK PLAN - CCTV ROUTING (CORP/PSNET) - 5TH FLOOR
MOC-C40-08-01-050000-F-01268132	IT NETWORK PLAN - CCTV ROUTING (CORP/PSNET) - BOILER
MOC4-C40-08-9600-CBL-027-AFRY	CABLE LIST - (DATA / CCTV / ACCESS CONTROL)
MOC4-C40-06-BOM-016-AFRY	BILL OF MATERIALS - POWER - CABLE DISTRIBUTION FLOOR
MOC4-C40-08-9606-EQL-025-AFRY	EQUIPMENT LIST - CCTV SYSTEM DEVICES - (CORP/PSNET)

- **SP – SPINE BUILDING (S1)**

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MOC-S1-08-01-050000-F-01202178	IT NETWORK PLAN - CCTV POINTS PLACEMENT - 1ST FLOOR
MOC-S1-08-01-050000-F-01202179	IT NETWORK PLAN - CCTV POINTS PLACEMENT - 2ND FLOOR
MOC-S1-08-01-050000-F-01202201	IT NETWORK PLAN - CCTV ROUTING (CORP/PSNET) - 1ST FLOOR
MOC-S1-08-01-050000-F-01202202	IT NETWORK PLAN - CCTV ROUTING (CORP/PSNET) - 2ND FLOOR
MOC4-S1-08-0500-CBL-002-AFRY	IT CABLE LIST

- AP – ASEPTIC PRODUCTION (A1)**

MOC-A1-06-DD-00-002020-F-01089299	BUILDING A1 – 1 ST FLOOR – CCTV, DATA AND VOICE PLAN – PART 1
MOC-A1-06-DD-00-002020-F-01089300	BUILDING A1 – 1 ST FLOOR – CCTV, DATA AND VOICE PLAN – PART 2
MOC-A1-06-DD-00-002020-F-01089301	BUILDING A1 – 1 ST FLOOR – CCTV, DATA AND VOICE PLAN – PART 3
MOC-A1-06-DD-00-002020-F-01089303	BUILDING A1 – 1 ST FLOOR – CCTV, DATA AND VOICE PLAN – PART 4
MOC-A1-06-DD-00-002030-F-01089306	BUILDING A1 – 2 ND FLOOR – CCTV, DATA AND VOICE PLAN – PART 1
MOC-A1-06-DD-00-002030-F-01089307	BUILDING A1 – 2 ND FLOOR – CCTV, DATA AND VOICE PLAN – PART 2
MOC-A1-06-DD-00-002030-F-01089311	BUILDING A1 – 2 ND FLOOR – CCTV, DATA AND VOICE PLAN – PART 3
MOC-A1-06-DD-00-002030-F-01089314	BUILDING A1 – 2 ND FLOOR – CCTV, DATA AND VOICE PLAN – PART 4
MOC-A1-06-DD-00-002031-F-01089315	BUILDING A1 - 2 ND FLOOR INTERSTITIAL - CCTV, DATA AND VOICE PLAN - PART 1
MOC-A1-06-DD-00-002031-F-01089317	BUILDING A1 – 2 ND FLOOR INTERSTITIAL - CCTV, DATA AND VOICE PLAN - PART 2
MOC-A1-06-DD-00-002031-F-01089319	BUILDING A1 – 2 ND FLOOR INTERSTITIAL - CCTV, DATA AND VOICE PLAN - PART 3
MOC-A1-06-DD-00-002031-F-01089323	BUILDING A1 – 2 ND FLOOR INTERSTITIAL - CCTV, DATA AND VOICE PLAN - PART 4
MOC-A1-06-DD-00-002040-F-01089324	BUILDING A1 - 3 RD FLOOR - CCTV, DATA AND VOICE PLAN - PART 1
MOC-A1-06-DD-00-002040-F-01089326	BUILDING A1 - 3 RD FLOOR - CCTV, DATA AND VOICE PLAN - PART 2
MOC-A1-06-DD-00-002040-F-01089327	BUILDING A1 - 3 RD FLOOR - CCTV, DATA AND VOICE PLAN - PART 3
MOC-A1-06-DD-00-002040-F-01089330	BUILDING A1 - 3 RD FLOOR - CCTV, DATA AND VOICE PLAN - PART 4
MOC-A1-06-DD-00-002041-F-01089335	BUILDING A1 – 4 TH FLOOR - CCTV, DATA AND VOICE PLAN - PART 1
MOC-A1-06-DD-00-002041-F-01089337	BUILDING A1 – 4 TH FLOOR - CCTV, DATA AND VOICE PLAN - PART 2
MOC-A1-06-DD-00-002041-F-01089338	BUILDING A1 – 4 TH FLOOR - CCTV, DATA AND VOICE PLAN - PART 3
MOC-A1-06-DD-00-002041-F-01089344	BUILDING A1 – 4 TH FLOOR - CCTV, DATA AND VOICE PLAN - PART 4
MOC-A1-06-DD-04-002600-F-01089219	BUILDING A1 - TELECOMMUNICATIONS ARCHITECTURE - CCTV, DATA AND VOICE

- EXTERNAL AREAS**

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MOC4-TS09-PGEN-06	FFEX – EXTERNAL AREAS TELECOM – CCTV TECHNICAL SPECIFICATION
MOC4-DAS03-PGEN-06	FFEX – EXTERNAL AREAS TELECOM – CCTV DATA SHEETS
MOC4-06-CBL-003-PGEN	FFEX – EXTERNAL AREAS TELECOM – CABLE LIST
MOC-SITE-06-01-000000-F-01066405	FFEX - EXTERNAL AREAS - TELECOM - NETWORK CABLE ROUTE PLANTS - PLAN 2/2
MOC-SITE-06-04-000000-F-01127551	FFEX - EXTERNAL AREAS - TELECOM - CCTV ARCHITECTURE DIAGRAM

Additional documents for the other areas mentioned will be provided by the CLIENT at a subsequent point.

6. ACRONYMS AND ABBREVIATIONS

• AWG	American Wire Gauge
• CAM	Cameras
• CAT 6A	Category 6A Cable
• CMOS	Complementary Metal-Oxide-Semiconductor
• DF	Distribution Location
• FPS	Frame per Second
• FTP	Foiled Twisted Pair
• Gbps	Gigabits per Second
• GHz	Gigahertz
• HDMI	High-Definition Multimedia Interface
• IDF	Access Location
• IP	Internet Protocol
• IP	Internet Protocol
• LAN	Local Area Network
• LCD	Liquid Crystal Display
• LED	Light-Emitting Diode;
• LSZH	Low Smoke Zero Halogen
• MDF	Main Distribution Location
• NTSC	National Television System Committee
• NVR	Network Video Recorder
• OSD	On Screen Display;
• PoE	Power Over Ethernet
• PTZ	Pan/Tilt/Zoom;
• PVC	Polyvinyl Chloride

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- QSFP Quad Small Form-factor Pluggable
- RFI Request for Information
- SFP Small Form-factor Pluggable
- SNMP Simple Network Management Protocol;
- SURVEILLANCE Security Cameras
- TA Tecnologia de Automação
- TCP Transmission Control Protocol
- TI Tecnologia da Informação
- TO Tecnologia de Operação
- UPS Uninterruptible Power Supply
- UTP Unshielded Twisted Pair
- Vac Volt Alternating Current
- VGA Video Graphics Array;
- VLAN Virtual Local Area Network;
- W Watts

Some acronyms and abbreviations not presented above may be defined when addressed in the texts described in this specification.

7. INSTRUCTIONS FOR PROPOSAL SUBMISSION

7.1 GENERAL

NOVO NORDISK requests the BIDDER, potential CONTRACTOR, to submit a detailed Technical and Commercial Proposal for the composition of the contracting of the object addressed herein, allowing the most rigorous evaluation possible. The proposal must strictly comply with the specifications in this tender and reference documents.

The proposals to be submitted must be based on the data, standards, certifications, and documents related to this tender. The BIDDER's experience in this type of service will also be considered, provided it is clearly indicated and evidenced in the technical proposal.

Any deviations from this specification must be clearly indicated in the proposal with due justification for non-compliance. Items not cited as deviations will be interpreted as accepted by the BIDDER and fully adhered to the specifications and standards of NOVO NORDISK.

NOVO NORDISK reserves the right, at its sole discretion, to reject any proposal or even cancel this bidding process, and under no circumstances will it be required to justify its decision to any BIDDER. It is understood that

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such actions will always be taken in favor of this contracting and without reimbursement of any expenses incurred by the BIDDER.

The BIDDER must use its knowledge and familiarity with other similar services already performed in this type of contracting to guide the working conditions to be developed within the stipulated period, following the best techniques and procedures for executing the proposed services, guaranteeing them against any technical failures, inappropriate standards, criteria, procedures, and specifications, assuming technical responsibility for their execution in accordance with current legislation.

The BIDDER may present alternatives that result in economic advantages for NOVO NORDISK, provided they offer equal or superior solutions to those described in NOVO NORDISK's internal standards. These alternatives must be presented separately for evaluation and approval.

The submission of the Technical and Commercial Proposals implies acceptance by the BIDDER of all conditions of this tender.

Once the technical proposals are received, all doubts are clarified, and the supply scope consolidated, after authorization from NOVO NORDISK through the issuance of the Technical Definition, only then shall the BIDDER submit the commercial proposals (via electronic means) exclusively to NOVO NORDISK/AFRY's Procurement department.

Additionally, the CLIENT reserves the right to:

- Reject any proposals, without the need to justify the reasons for such a decision;
- Not accept proposals delivered after the established date;
- Change, delete, or add, at any time, information, conditions, dates, and deadlines contained in this Request for Proposal and its Attachments;
- Suspend, revoke, or annul this request for proposals at any time;
- Include or exclude BIDDERS at any time;
- Choose and decide on the proposals that best serve their interests.

7.2 CLARIFICATIONS AND QUESTIONS

Requests for additional information must be submitted in writing in Portuguese (only via email) up to three business days before the final date for proposal submission, referencing: Tender No. **MOC4 -08-9600-RFQ-004-AFRY – RFQ – CONSTRUCTION AND CONFIGURATION OF CCTV SYSTEM.**

Clarification requests regarding the bidding process must be sent to the Procurement area in interface with the CLIENT's technical responsible. The BIDDER must promptly acknowledge in writing (email) that they have received this information and that it is being considered in the preparation of the proposal.

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7.3 CONFIDENTIALITY

BIDDERS must not disclose to third parties, except those directly involved in preparing their proposals, any details of the bidding or invite unauthorized companies to participate formally by NOVO NORDISK. This restriction remains in effect, even for non-selected BIDDERS. The BIDDER acknowledges that, while responding to this inquiry and providing services to NOVO NORDISK, certain information may be disclosed to the SUPPLIER that is confidential and proprietary to NOVO NORDISK ("Confidential Information").

The SUPPLIER agrees that neither it nor any of its directors, employees, or agents shall, directly or indirectly, disclose to third parties any confidential information provided by or on behalf of NOVO NORDISK or use such information for any purpose other than responding to this inquiry or providing services to NOVO NORDISK. The SUPPLIER must take all necessary measures to maintain the confidentiality of the information in its possession and, at the conclusion of this bidding process, all documents received by the BIDDER used to prepare the proposal must be destroyed/deleted/eliminated.

7.4 GENERAL PROPOSAL REQUIREMENTS

The BIDDER must consider and prepare its proposal so that its scope complies with this tender and other reference documents such as: Detailed Designs, Technical Specifications, Service Spreadsheets, Equipment and Materials Lists, Drawings, Diagrams, and others.

If the BIDDER is not the supplier of one or more system components required, it must be responsible for the integration of all teams and supplies, remaining accountable for the guarantee of the entire system.

The BIDDER is responsible for consulting the documents referenced in this specification to estimate the equipment and materials to be supplied.

7.5 SUPPLY SCOPE

Based on this specification, the BIDDER is requested to submit a proposal for supplying a complete package for "DEVELOPMENT AND CONFIGURATION OF CCTV SYSTEM" including including accessories, calibration, configuration, installation materials, coordination services, design, testing, assembly, installation, licensing, updates, permissions, configuration, programming, alarm monitoring, display creation/configuration, third party integrations (Access Control System), backups, testing (functional, acceptance, backup/retrieval, etc.) and operator training, within the limits established in this document.

All auxiliary equipment and systems within the supply limits and essential for the proper functioning of the scope of this specification must be considered.

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7.6 INSTRUCTIONS FOR TECHNICAL PROPOSAL CONTENT

The Technical Proposal must include, at a minimum, all the requirements described in the following items and sub-items. It is emphasized that the content presented should not be strictly limited to these topics, and it is highly recommended to include additional information that enriches and demonstrates full understanding and adherence to the project requirements.

7.6.1 TECHNICAL PROPOSAL

It is mandatory to present in the original technical proposal all technical specifications, catalogs, functionalities, and any other relevant information regarding the proposed equipment, devices, and materials, including model, "part number," and manufacturer's name. The OWNER's "VENDOR LIST" must be followed.

The same applies to the described services to be performed, where the BIDDER must clearly state in its technical proposal the scope, execution sequence, description, deadlines, and any other relevant information for the analysis, qualification, and potential to carry out the requested work.

The BIDDER must comply with all applicable laws, standards, and regulations, not limited to local and national regulatory bodies for the project's installation location.

All BIDDERS must strictly follow the following content structure/items when submitting the technical proposal:

- Cover;
- Company name;
- Address;
- Document title / Number / Revision;
- Date and location;
- Proposal validity;
- Introduction;
- Recent supplies with similar characteristics, including buying company name and contact person;
- Company certificates pertinent to the project scope;
- Brief company description, with a maximum of 3 paragraphs, containing:
 - Company size, number of employees;
 - Size of technical support in Brazil;
 - Installed base in industrial plants in Brazil;
 - Market share with the offered products.

The technical proposals must contain a complete and detailed description of the scope, listing all supplies and services to be executed in accordance with this Tender.

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A clear list of exclusions and deviations (elements that are handled differently from what is written) from this Tender must be provided at the end of this item.

7.6.2 TECHNICAL SOLUTION

Technical description of the solution offered for each system, including drawings with the complete architecture of the system(s).

7.6.3 SERVICES, MATERIALS, AND EQUIPMENT SPREADSHEET

The BIDDER must present spreadsheets pertaining to services, materials, and equipment, separately, duly completed with quantities, without prices, and reflecting all items necessary for the full fulfillment of the scope.

The BIDDER, as the potential CONTRACTOR, is entirely responsible for the spreadsheet submitted in its proposal, as it must verify and confirm all provided information, indicate the quantities for each item, and correct any discrepancies based on the project documentation, information from the request for proposal, and clarifications made during the bidding period.

The BIDDER must include in its proposal a spreadsheet with an item called "OMITTED," where all services and materials that may not have been mentioned in this specification but are necessary for the total and perfect execution of the scope implementation must be presented.

7.6.4 SUPPLY SCHEDULE

The BIDDER must present a preliminary schedule of the services related to the contractual object, preferably using MS Project software or similar (such as Primavera P6). Climatic restrictions, weekends, holidays, and other regional commemorative dates must have their respective impacts considered in the schedule's detailing and must be detailed according to the Phases that the Project will have, as described in this document.

The schedule milestones must be related to the respective activities that define them.

The BIDDER must at least present the following "Milestones/Stages" in the preliminary schedule:

- Planning;
- Detailed Engineering;
- Procurement, Acquisition, and Delivery;
- IN-HOUSE Assembly (if applicable);
- Mobilization of Own and Third-Party Teams;
- On-site Assembly and Installations;
- Startup with Provisional Acceptance;
- Final Acceptance with As-Built documentation.

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An illustrative template is provided for the BIDDER to consider for the presentation and preparation of its preliminary schedule.

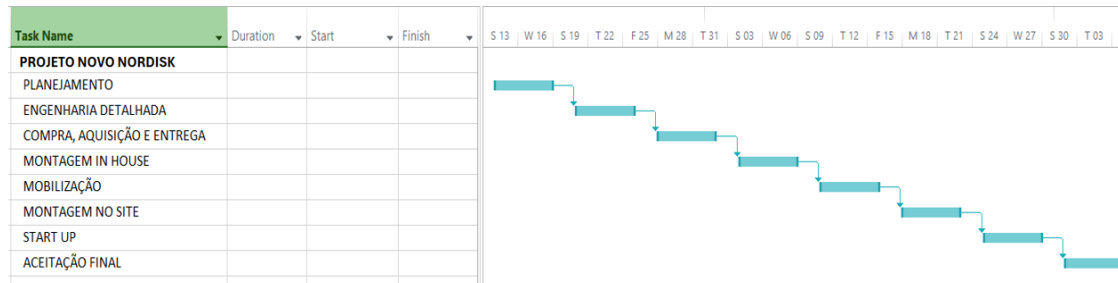


Figure 1 – Illustrative Template

The BIDDER must outline the dates associated with its solution strategy, assuming it becomes the CONTRACTOR. If necessary, the processes mentioned here may be further detailed later between the OWNER's and CONTRACTOR's planners.

7.6.5 ORGANIZATIONAL CHART AND HISTOGRAM

The BIDDER must present the proposed Organizational Chart for the development of the services covered by this reference tender, showing its linkage with the company's headquarters organization and the respective Responsibility Matrix.

The BIDDER must also present, in the form of a histogram (Direct and Indirect Resources) for the entire contractual period, the quantities planned month by month for the services inherent to the requested scope.

Additionally, the BIDDER must include in its proposal the list and histogram of all equipment that will be mobilized for the execution of the assembly services covered by this tender, also informing whether the equipment is owned or rented/third-party.

7.6.6 LIST OF SUBCONTRACTORS

The CONTRACTOR may subcontract partners for the supply and installation of hardware/software and materials/installations, provided that the responsibility and management over the supply remain exclusively with the CONTRACTOR. The BIDDER must clearly state its possible intention to subcontract, explaining the nature of these and committing to submitting them for approval by the OWNER.

A preliminary list of subcontractors must be presented along with the proposal. If the BIDDER does not intend to subcontract, this intention must be declared in the proposal.

If the BIDDER is not the supplier of one or more required system components, it must formally confirm that it will be responsible for the integration of all teams and supplies, remaining as the sole party responsible for the entire System's guarantee.

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7.6.7 COMPANY TECHNICAL QUALIFICATION

The BIDDER must include in the technical proposal a list of similar projects already supplied in the industrial area, accompanied by all documents that prove the execution of the listed scope. The list must specify, at a minimum:

- Client name and contact details;
- A summary of the applied solution;
- Execution date.

Additionally, the BIDDER must:

- Present technical capacity certificates compatible with the marketing and execution of services referenced in this tender;
- Have qualified technical professionals as part of the company's technical team.

7.6.8 PROPOSAL VALIDITY

The proposal must be valid for a minimum period of 90 (ninety) days from the date of delivery of the last version after the technical alignment meeting(s).

7.7 INSTRUCTIONS FOR COMMERCIAL PROPOSAL CONTENT

7.7.1 COMMERCIAL PROPOSAL

The Commercial Proposal must not contain technical items or exclusions; it should be limited to commercial items, such as:

- Offered unit prices;
- Social and labor charges;
- General expenses;
- Considered taxes;
- Expected disbursement schedule;
- Base date.

7.7.2 COMMERCIAL CONTRACTING MODALITY

This package will be contracted on a lump-sum basis, in accordance with the content of this specification.

Clarifications or additional commercial information must be made by writing via email, EXCLUSIVELY, to a member of the procurement team.

It is emphasized that commercial matters should only be directed to the procurement team member, and no other NOVO NORDISK team member should be involved in commercial discussions.

Prices must be presented considering the following conditions:

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- Lump sum for the entire scope (monthly payment for equipment rental);
- All taxes included;
- General services;
- Spare parts;
- Optional items;
- Imported and domestic equipment;
- Installation materials (application and consumption);
- Necessary software and licenses;
- Engineering services;
- Manufacturing and assembly services;
- Supervision services;
- Assisted operation, commissioning, and startup;
- Support, operation, and maintenance services;
- Training.

Included costs in the supply:

- Travel and accommodation expenses (including visa);
- Per diem and meals for technical personnel;
- Car rental and/or other transportation expenses;
- Federal and local fees;
- Communication expenses such as phone calls, internet links, etc.

The BIDDER must calculate and include in its proposal the FIXED UNIT PRICE for additional materials and equipment for the system in question requested up until the time of its demobilization (addressing the supply of installation materials, equipment, and necessary services).

Other incidental services will only be paid if accompanied by a specific request from the OWNER, a specific proposal from the CONTRACTOR, and expressing approval by the OWNER for the execution of such services. These must include, at a minimum, the service description, the quantities involved, technical justification for its necessity, and its start and end dates.

8. OVERVIEW OF THE PROJECT

The BIDDER must consider, within their supply scope, all services, materials, and equipment necessary for the implementation of the entire structured cable infrastructure for the following buildings:

- **C40** – Utilities Building and its respective floors.

Fill & Finish Expansions building the future for patients	Document Title	Rev.	Project FFEx MoC	
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Figura 2 – Indication of the Utilities Building – C40

- **C40** – Boiler.

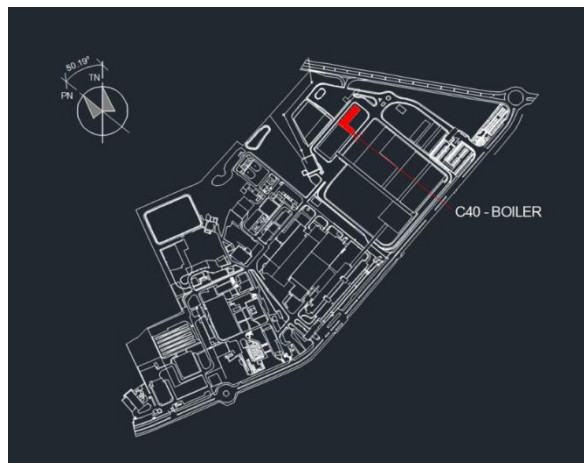
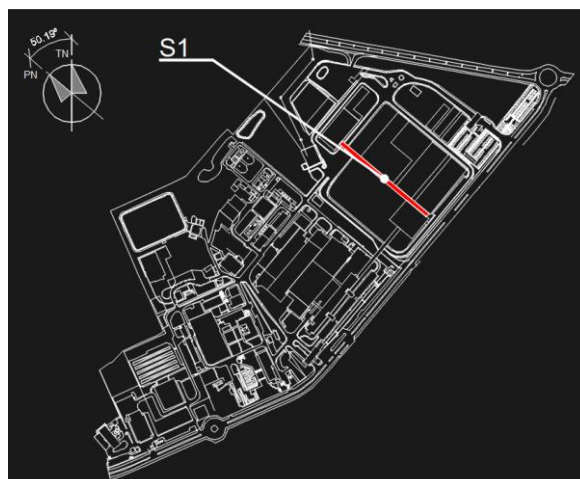


Figura 3 – Indication of the Boiler – C40

- **S1** – Spine Building and its respective floors.



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Figura 4 – Indication of the Spine – S1

- **AP** – AP building and its respective floors
- **External areas**

The BIDDER must also provide the necessary cameras for the external areas.

- **D20 and D21** – Datacenters.

The data centers already have an existing CCTV system. This system includes a total of 22 cameras (11 per datacenter being 5 internal and 6 external). The bidder must integrate these existing cameras into the new system.

NOTE: Cameras are included in the Utility Building, Boiler and Spine Building in this project. All devices allocations must be made in accordance with the project documents (drawings, bill of materials, equipment specifications, etc.).

9. COMPOSITION OF THE CCTV SYSTEM

The main specifications for the CCTV Systems are based on the guidelines defined by NOVO NORDISK. See REFERENCES DOCUMENTS.

The objective of this tender is to clearly describe how Novo Nordisk's global network is structured to ensure both functional efficiency and robust security.

The data network system is responsible for managing corporate data traffic, including applications related to data, voice, and video, ensuring high-performance connectivity across various segments of the plant's infrastructure.

Although the CCTV systems share the same physical infrastructure with the corporate data network, they must remain logically segregated through the implementation of dedicated VLANs.

The CCTV systems, covering SECURITY applications, include the configuration of all security cameras within the buildings, ensuring effective surveillance and protection of all monitored areas.

All cameras must be properly licensed, certified, and fully integrated with the existing Video Management System (VMS) and storage hardware, which are already operational and certified within the Novo Nordisk facility.

This guarantees the reliable performance of the system and supports future scalability, including the integration of additional monitoring cameras as needed.

The security system will utilize the same network switches as the CORP and PSNET data networks. These switches will be installed within the DFs/IDFs racks. Despite sharing the same physical switches, the CORP and PSNET networks must remain logically segregated by VLANs.

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Any CCTV additional system proposed for corporate security must have its model and manufacturer pre-approved by Novo Nordisk.

The system must incorporate current technology, with a minimum of two (2) years of proven availability in the market, and must be designed for continuous, uninterrupted operation. All components subject to regulatory compliance must be certified and approved by Anatel.

The system shall be provided with all necessary resources and components required for its complete installation and operation.

9.1 GENERAL DESCRIPTION OF CCTV SYSTEM AND DEVICES

The system will be operated, supervised and managed through the control center operating stations in the existing control center at the industrial plant.

The following characteristics shall be considered:

- IP video cameras;
- i-VMD (Intelligent Video Motion Detection):
 - Motion detection;
 - Tamper detection through obstruction or deliberate repositioning;
 - Object detection, identifying removed or abandoned items and generating corresponding events, which must be recognized by the Video Management System.
- Media converter, if required;
- All special cables, terminals, connectors, accessories and other internal and external fittings required for installation shall be provided;
- For outdoor installations, all equipment must be housed in metal enclosures (weatherproof and waterproof);
- Providing adequate protection based on the installation environment;
- Support automatic restart after any type of communication failure;
- Facilitate detection and identification of system faults;
- Trigger alarms in the event of issues such as:
 - Power loss;
 - Other critical failures.

A closed-circuit television (CCTV) system is a crucial component of Novo Nordisk's security infrastructure, providing real-time information, surveillance, and monitoring capabilities. It is essential for preventing unauthorized access, verifying alarms, and collecting evidence to support security operations. An efficient CCTV system enhances security through proactive monitoring and timely incident response.

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Centralizing CCTV operations enables seamless integration and real-time data flow between systems, which is essential for effective monitoring and response. A centralized CCTV system addresses these challenges by offering a unified approach to security management, reducing the risk of communication failures, and ensuring faster, coordinated responses to incidents.

The main objectives of CCTV at Novo Nordisk are:

- Enhance security posture: Implement a centralized CCTV monitoring system that improves the ability to detect, assess, and respond to security incidents in real time.
- Continuous surveillance: Ensure continuous video surveillance of critical areas to prevent unauthorized access and minimize security risks at Novo Nordisk facilities.
- Standardize CCTV practices: Establish consistent CCTV protocols and procedures aligned with global best practices and organizational standards for video surveillance.
- Ensure compliance: Ensure all CCTV operations comply with the specific security requirements outlined in the Minimum Security Requirements (MSR) and AME GLP-1 CCTV Guidelines.
- Improve CCTV operational efficiency: Optimize the operation and management of the CCTV system to reduce downtime, improve video quality, and enhance response times during security incidents.
- Support organizational growth: Design a scalable CCTV infrastructure that can adapt to the growing needs of Novo Nordisk's global operations.

9.2 SPECIFIC DESCRIPTION OF CCTV DEVICES

9.2.1 FIXED CAMERA (CORP-SURVEILLANCE / PSNET-PROCESS) - BULLET (INDOOR/OUTDOOR)

Fixed BULLET cameras must be compatible with the central management system already in place at the plant and have the following minimum characteristics:

- General Features:
 - Outdoor bullet network camera designed for harsh environments
 - 4/3" progressive scan RGB CMOS sensor with ultra-high light sensitivity (Lightfinder 2.0)
 - Effective resolution of 10 MP (3712 × 2784), 4K UHD video up to 60 fps
 - Forensic WDR, Lightfinder 2.0, and OptimizedIR for outstanding low-light performance
 - Axis Object Analytics with deep learning-based object classification (humans, vehicles)
 - Audio input: 1× 3.5 mm TS mono connector
 - Local edge storage with microSD/microSDHC/microSDXC support and 8192 MB flash memory
 - PoE-out port for powering an external device (e.g., speaker, strobe, or another camera)

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- Axis Edge Vault for secure key storage and hardware protection (FIPS 140-2 level 2 certified)
- Certifications: IP66/IP67, IK10 (housing), IK08 (window), NEMA 4X, UL/cUL, BIS, UKCA, CE, KC, EAC, VCCI, RCM
- TAA compliant
- Field of View:
 - Varifocal Canon lens: 12–48 mm (optional 50–150 mm) – 4× optical zoom (3× for telephoto)
 - Horizontal field of view: 90°–21° (or 21°–7° for 150 mm model)
 - Vertical field of view: 49°–12° (or 12°–4° for telephoto model)
- Minimum Illumination:
 - Color: 0.02 lux (with Lightfinder 2.0)
 - B/W: 0 lux with IR illumination on
- Day/Night Operation:
 - Automatic removable IR-cut filter for day/night functionality
- Infrared (IR):
 - Integrated IR LEDs (850 nm wavelength)
 - IR illumination range: up to 100 m (328 ft), or up to 120 m (394 ft) on 150 mm model
- Video Compression:
 - H.264 (Baseline, Main, High profiles)
 - H.265 (HEVC)
 - Motion JPEG
 - Axis Zipstream (optimized for storage and bandwidth efficiency)
- Video Streams:
 - Up to 60 fps in 4K resolution
 - Supports up to 20 individually configurable video streams
- Electronic Image Stabilization (EIS):
 - Yes, built-in electronic image stabilization
- Wide Dynamic Range (WDR):
 - Forensic WDR with Lightfinder 2.0
 - Designed for excellent detail in scenes with both dark and bright areas
- Backlight Compensation:
 - Supported via Forensic WDR technology
- Automatic Features:

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- Automatic Gain Control (AGC)
- Automatic White Balance (AWB)
- Automatic Exposure Control (AEC)
- Lens distortion correction, barrel correction, defog, adaptive compression
- Scene profiles (Forensic, Vivid, Traffic Overview)
- Auto day/night switching, image rotation (0°, 90°, 180°, 270°), privacy masking, overlay text/graphics
- Shutter Speed Range:
 - From 1/45500 s to 2 s (depending on scene and exposure settings)
- Network Interface:
 - 1 × RJ45 Gigabit Ethernet (PoE+ Class 4 or PoE++ Class 6)
 - PoE-out support for external devices
- Standards and Protocols:
 - IPv4/v6, HTTP, HTTPS, SSH, SSL/TLS, IEEE 802.1X, SNMP v1/v2c/v3, RTSP, RTP, SRTP, FTP, SFTP, SMTP, LLDP, CDP, IGMP v1–3, DNS, DHCP v4/v6, NTP, MQTT, DNS/Bonjour, QoS
 - ONVIF Profiles G, M, S, T
 - Syslog, usgV6, and more
- Power Supply:
 - PoE IEEE 802.3at Class 4 (typical: ~13.7 W / max: 25.9 W)
 - PoE IEEE 802.3bt Class 6 (typical: ~14.9 W / max: 51 W when using PoE-out)
 - Optional DC input: 10–28 VDC (typical: ~13.7 W / max: 25.9 W)
- Operating Temperature:
 - From -40 °C to 60 °C (up to 74 °C per NEMA TS 2)
 - Humidity (operation): 10–100% RH, condensing
 - Humidity (storage): 5–95% RH, non-condensing
- Enclosure Protection:
 - Housing: Aluminum body with IK10 vandal-resistant rating
 - Window: IK08 impact protection
 - Weather-resistant: IP66 and IP67 rated
 - NEMA 4X protection for harsh outdoor environments
- Mounting Options:
 - Wall or pole mounting (default)

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- Optional accessories for ceiling or column mounting
- Repaintable surface for architectural integration

➤ **VENDOR LIST – NOVO NORDISK REFERENCE**

✓ **Manufacturer: Axis Communications**

▪ **BULLET CAMERA (INDOOR/OUTDOOR)**

- **Model: AXIS Q1808-LE**
- **Part Number (P/N): 02507-001**

9.2.2 **FIXED CAMERA (CORP-SURVEILLANCE / PSNET-PROCESS) - DOME (INDOOR)**

Fixed dome cameras must be compatible with the central management system already in place at the plant and have the following minimum characteristics:

- General Features;
 - Ultra-compact, fixed mini dome network camera;
 - 8 MP resolution (3840 x 2160);
 - 1/2.8" progressive scan RGB CMOS sensor;
 - Fixed lens: 2.9 mm, F2.0, IR-corrected;
 - Pre-installed analytics: AXIS Object Analytics, Scene Metadata, Live Privacy Shield, Video Motion Detection (VMD), Tampering Alarm, Audio Detection;
 - Deep Learning Processing Unit (DLPU) for advanced analytics on the edge;
 - Cybersecurity features: Signed OS, Secure boot, Axis Edge Vault with secure element;
 - Local storage: microSD/microSDHC/microSDXC (encrypted storage supported).
- Field of View:
 - Horizontal: 109°
 - Vertical: 56°
- Minimum Illumination:
 - Color: 0.25 lux at 50 IRE, F2.0
 - B/W: 0.05 lux at 50 IRE, F2.0
- Day/Night Operation:
 - True day/night functionality with an automatic IR-cut filter.
- Infrared (IR):
 - IR-corrected optics.
- Video Compression
 - H.264;

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- H.265;
- MJPEG;
- Zipstream technology for reduced bandwidth and storage usage;
- Video Streams: Multiple simultaneous streams;
- Wide Dynamic Range (WDR): Available;
- Backlight Compensation: Yes;
- Automatic Features:
 - Auto white balance (AWB);
 - Auto gain control (AGC);
 - Auto exposure;
 - Pre-installed video analytics for automatic detection and classification of humans and vehicles.
- Network Interface:
 - 10/100BASE-T Ethernet;
 - RJ45 connector.
- Standards and Protocols:
 - ONVIF Profile S;
 - ONVIF Profile G (recording and playback);
 - IPv4, IPv6;
 - RTSP, HTTP, HTTPS;
 - SNMP, NTP, DNS, DDNS, FTP, SMTP, DHCP.
- Power Supply:
 - Power over Ethernet (PoE) IEEE 802.3af/802.3at Type 1 Class 2.
 - Typical Power Consumption: 4.2W (maximum).
- Operating Temperature:
 - 0 °C to 40 °C (32 °F to 104 °F)
- Enclosure Protection:
 - IP42 rated for dust and dripping water protection;
 - IK08: Vandal resistance (protection against 5 joules impact).
- Mounting Options:
 - Surface mount (ceiling or wall);
 - Optional mounting accessories available for pendant or recessed installations.

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➤ **VENDOR LIST - NOVO NORDISK REFERENCE**

✓ **Manufacturer: Axis Communications**

▪ **DOME CAMERA**

- **Model: AXIS M3088-V**
- **Part Number: 02375-001**

NOTE: All camera license quantities required for integration into the image management and recording system must be provided by the BIDDER, as well as all necessary configurations and adjustments to the existing system for the new cameras.

9.2.3 VIDEO MANAGEMENT SOFTWARE

The video management system will be distributed across all buildings of FFEx MOC, including administrative, operational, production, technical, and support areas, as well as external circulation zones and the perimeter. The CCTV system will be **Avigilon Control Center**, in compliance with the standardization adopted by SITE MOC's Asset Security.

The CCTV system will function as a technological surveillance barrier, with cameras strategically positioned to cover critical points, entrances, exits, and risk areas.

The barriers include:

- Continuous 24/7 monitoring.
- High-resolution recording with secure storage.
- Motion detection and automatic alerts.

The CCTV system will be integrated with the access control and intrusion alarm systems, enabling:

- Correlation of events between people's entry and captured images.
- Automatic activation of recording in case of physical barrier violation.
- Coordinated response among systems during emergencies or unauthorized access attempts.

9.2.4 NVR (NETWORK VIDEO RECORD)

The supply scope of the BIDDER does not include the creation or maintenance of NVRs as this will be done by the OWNER. The BIDDER must however inform NovoNordisk's IT team of the storage specification for the video recordings and backup according to the OWNER's requirements as well as any other necessary requirements.

9.2.5 OPERATION STATIONS

The BIDDER must configure the operation stations that the OWNER will supply for the proper image monitoring system requirements.

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9.3 Infrastructure for cable accommodation (Metallic and Optical)

All infrastructure was described in the document **(MOC4-C40-08-9600-ES-009-AFRY - TECHNICAL SPECIFICATION STRUCTURED CABLING (CORP/PSNET) - DATA, VOICE, VIDEO AND WI-FI)**.

For the installation of the project cameras, it is within the BIDDER's scope to perform minor infrastructure adjustments (when necessary) to ensure that the camera installations meet the quality standards required by the project.

9.3.1 ELECTRICAL POWER SUPPLY

All camera power supplies will be provided through their respective switch ports using PoE technology.

9.4 Functional Requirements for the CCTV System Implementation

Below are the main functional requirements that must be followed for the CCTV System Implementation. For the full list of requirements the BIDDER must refer to the reference document MOC4-25-0962-ES-001. All requirements listed in this document must be followed:

- **Integration with other security systems:** Integration with access control systems, intrusion alarm systems, perimeter detection systems, the security operations center, and the gate perimeter system.
- **Scalability:** The system must be designed to support future expansions, allowing the addition of more cameras and updated technology without requiring a complete overhaul of the existing infrastructure.
- **Testing and commissioning:** The CCTV system design must include plans for complete testing and commissioning to verify that all components are functioning correctly.
- **Documentation:** All design work must be meticulously documented, including detailed drawings, system architecture, and specifications. This documentation must be kept updated and available for review during audits and inspections.
- **Intuitive user interfaces:** User-friendly interfaces that provide CCTV operators with easy access to critical information and controls. The interface must be simple and intuitive, allowing operators to quickly navigate between camera feeds and control settings.
- **Customizable dashboards:** Dashboards must allow operators to customize layouts to prioritize specific camera feeds, data streams, and alerts, improving situational awareness and enabling more effective responses to incidents.
- **Video retention:** Video recordings must be stored for at least 30 days.
- **Automatic deletion:** All video recordings must be automatically deleted after the retention period, unless extended based on a valid justification (e.g., security incidents requiring longer retention).

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- **Redundancy and failover:** To ensure reliability and redundancy, dual-server configurations are required, with automatic failover between two server rooms. In the event of a server failure, camera recordings must automatically switch to the backup server within minutes, ensuring minimal interruption (no more than 5 minutes of recording loss).
- **Alarm integration:** IAS alarm signals must be transferred to the CCTV system as described in IAS standards, where the IAS central unit communicates through the IT network. Preset positions for alarm points must be configured so that cameras automatically move to these locations when an alarm is triggered. These presets must be defined and agreed upon during the detailed design phase to ensure comprehensive coverage and rapid response.
- **VMS platform capabilities:** The VMS platform must support advanced features such as scalable architecture, high-resolution video support, and reliable performance under high load. The platform must offer robust reporting and analytical tools to support security operations and decision-making.
- **Active Directory integration:** The VMS platform must integrate with Active Directory.
- **Scalability and robustness:** The VMS platform must be robust and scalable to support future expansions and upgrades.
- **User roles and access levels:** The VMS must allow user roles and access levels to ensure that only authorized personnel can view or manage specific camera feeds, maintaining strict access control.
- **Strong password policies:** The VMS must enforce strong password policies, including regular updates and complexity requirements, to enhance security.
- **Comprehensive logging:** The VMS must implement a comprehensive log system to track access and system changes for auditing and security purposes, ensuring accountability and traceability.
- **Camera view configuration:** The VMS must allow configuration of camera views, including field of view and preset positions, to ensure optimal coverage and rapid incident response.
- **Intuitive and customizable interface:** The VMS must allow configuration of an intuitive and easy-to-use interface with customizable dashboards for efficient monitoring and management. The graphical user interface (GUI) must enable security personnel to quickly access and control camera feeds and system settings.
- **Advanced AI-based analytics:** The VMS must support advanced AI analytics such as motion detection, facial recognition, and behavior analysis, used to enhance incident detection and response capabilities.
- **Data security:** Ensure end-to-end encryption, regular updates, and compliance with ONVIF standards to maintain data protection and system integrity.

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- **Technical alarm notifications:** Ensure the VMS can send technical alarms to a ServiceDesk for immediate attention and resolution of system issues, helping maintain system reliability and performance.
- **Incident management module (if possible):** If available, implement an incident management module within the VMS that allows operators to register, track, and manage incidents directly from the system, streamlining responses and ensuring consistent documentation.
- **Compliance with privacy laws:** The VMS platform must comply with all relevant privacy laws and data protection regulations. Implement privacy zones and masking features on camera feeds when necessary to protect sensitive areas.
- **Search tools:** The VMS must provide search tools for motion events and date/time searches.
- **PTZ control:** The VMS must support PTZ camera control via mouse or plug-in joystick.
- **Network configuration:** Ensure the SAN is integrated into a high-performance network, typically through fiber-optic technology.
- **Gatehouse CCTV design:** Entry-lane CCTV must be designed for both gatehouse security monitors and SOC (Security Operations Center) monitors.
- **Fence-line CCTV:** CCTV covering the fence line must be connected to the PDS. Video analytics may be used as a replacement or complement to a dedicated PDS.
- **CCTV integration with IAS/Access Control:** It is preferable for the CCTV system to be integrated with IAS/Access Control (EcoStruxure Security Expert), enabling quick verification in case of alarms, minimizing response time, and supporting guards and the Alarm Center in notifying law enforcement.
- **Internal viewing monitors:** To allow staff inside the building to observe external conditions before opening doors/gates, external CCTV feeds should be displayed on a monitor (TV screen) installed indoors.
- **High-risk area recording:** Cameras covering high-risk areas (e.g., third defense layer or product-flow surveillance) must record continuously. Other cameras may begin recording upon motion activation. In all cases, recording must be at least 12 fps, and image quality must meet its intended purpose (e.g., verification or identification)

10. SPARE PARTS

The BIDDER must provide a list of recommended spare parts for continuous operation over a period of **2 years**, considering passive items and equipment.

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11. SERVICES

The bidder must include the following technical services in their proposal, including the management and organization of each active team:

- **Installation of the Video Management Software:** This step includes deploying and configuring the CCTV video management platform on the designated servers or workstations. It ensures the system is properly installed, operational, and ready to receive camera streams and user configurations. This installation must be done under supervision of NovoNordisk's IT department.
- **Implementation of Licensing and Firmware Updates:** In this phase, all necessary software licenses must be activated, and camera firmware or system firmware must be updated to the latest supported versions. This guarantees compliance, stability, and access to the newest security features.
- **User and permission structure configuration:** Creation and configuration of operator accounts, administrative roles, access groups, and permission levels, ensuring that system privileges align with organizational policies and operational requirements.
- **Implementation of Cameras in the Systems:** This involves adding each camera to the management system, configuring IP addresses, streaming parameters, and assigning them to corresponding locations or maps. It ensures all video devices are recognized and integrated. This also includes programming of any necessary logic for the proper functioning of the system.
- **Integration of Existing Cameras into the Systems:** This involves adding and configuring the existing cameras of the Data Centers (D20 and D21) into the new video monitoring system. This must include any necessary licensing and communication as well as adding the cameras to the system, configuring IP addresses, streaming parameters, and assigning them to corresponding locations or maps. Each Datacenter has 5 internal cameras, 6 external cameras and 3 access control doors which must be integrated according to the item "Third Party Systems Integration" on this list.
- **Alarm monitoring:** Setup of alarm conditions where necessary.
- **Implementation and Configuration of Operator Workstation Displays:** This step includes setting up monitoring screens, video walls, or operator consoles, with layouts, view cycles, alarms, and permissions adjusted according to operational needs.
- **Implementation of Camera Analytics when Necessary:** Advanced video analytics—such as motion detection, intrusion detection, LPR, or object classification—are configured for cameras that require these features. This enhances security capabilities and enables automated event detection.
- **Third party system integrations:** Configuration and validation of integrations with external systems such as Access Control System for event association, or client-provided applications. The Access Control integration must include the relevant licensing for receiving and reacting to triggers from the Access Control System (EcoStruxure Security Expert) for specific monitoring conditions. The BIDDER must decide, implement and configure the communication protocols used for third party communications.

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The number of cameras integrated into the system shall be the same as the total number of doors monitored by the access control system.

- **Implementation of Image Backup System:** A backup mechanism is deployed to ensure recorded footage is securely duplicated, whether to secondary storage, cloud services, or long-term archival systems. This guarantees redundancy and preserves evidence.
- **Testing:** General functional tests are performed to validate image quality, camera connectivity, system performance, and behavior under normal and stress conditions. The goal is to confirm that the system meets operational requirements.
- **Recording, Storage and Image Retrieval Tests:** These tests validate the complete video lifecycle: recording continuity, storage integrity, retention time compliance, and successful retrieval of footage for investigation. It ensures the system performs reliably in real-world scenarios.
- **Operator and administrator training:** Training sessions for system operators and administrators covering system operation, camera management, alarm handling, reporting, and maintenance procedures as well as any other system functionalities included in this scope.

The BIDDER must also include the following services in its proposal, including the management and organization of each active team:

- Coordination meetings at the OWNER's premises during the project phase;
- On-site project coordination during the assembly, installation, configuration, testing, startup, and assisted operation phases;
- Coordination of interfaces with suppliers and subcontractors (Construction Company, Designer, and Third Parties);
- Detailed executive schedule for each project phase;
- Executive designs for system implementation;
- Transportation of equipment to the site, including all transportation and insurance costs;
- Storage of equipment on-site until installation and system startup;
- Documentation of equipment, system architecture, equipment configurations, scripts, interfaces, customizations, etc.;
- Documentation of system architecture, equipment configurations, scripts, interfaces, customizations, etc.;
- System documentation and self-documentation (manuals, data books, etc.);
- Installation documents;
- Field Acceptance Tests (FAT) after energization and interconnection of the entire system;
- Monitoring system startup and assisted operation (minimum of 30 days available);
- Performance testing and system certification;

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- “As-Built” documentation.

For all field services, the CONTRACTOR must include associated costs for air travel, transfers, hotel expenses, transportation to the site, meals, phone calls, PPE, and other costs inherent to the activity.

The BIDDER will be responsible for other general or legal expenses that directly impact on the service costs, such as:

- Service administration, engineers, and technicians;
- External and internal transportation (vertical and horizontal);
- Miscellaneous expenses such as emergency medications, office and cleaning materials for the work areas, quality control of applied materials, etc.

The CONTRACTOR must consider shipping the equipment and materials on more than one transport and on different dates.

The CONTRACTOR will be responsible for internal transportation within the site, storage, unpacking, and installation of equipment and other accessories at each location.

The CONTRACTOR must include in the scope the construction of an office (or the use of containers) within the site for the accommodation of its personnel and storage of equipment and materials (tools, materials, and equipment that will compose the offered systems), including security and cleaning of the site office, and compliance with the OWNER's site construction standards, to be provided at a later time.

The BIDDER must indicate in its proposal the estimated area required for its site facilities, as well as submitting a preliminary layout of the same.

The OWNER will only provide utilities (water, sewage, and power), with the CONTRACTOR responsible for connecting these utilities to its site facilities. The voltage step-down, provision of diesel generators, and contracting of telephone lines and data links, where necessary, will be at the CONTRACTOR's expense.

The BIDDER must submit an organizational chart of the team that will be part of the project phase and the field service phase, indicating the names of the main responsible persons and attaching their respective résumés.

The BIDDER must perform all equipment installation services under the supervision of an engineer registered with the CREA (Regional Council of Engineering and Agronomy). This includes the collection of ART(s) (Technical Responsibility Annotations) in Mato Grosso do Sul, and the opening of a tax registration in the municipality where the project will be executed for the collection of local ISS (Service Tax).

The BIDDER must comply, during the execution of services and installations, with the site's safety rules and standards.

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12. PROJECT DOCUMENTATION

The BIDDER must consider that the supply of all documents and manuals will be electronic, uploaded to the project repository for all phases of the work.

The OWNER will approve the formatting, layout, and content of all documents and drawings submitted by the CONTRACTOR. Although approval and release by the OWNER are required, all responsibility for the accuracy and completeness of the documents, including for the operation of the equipment, will rest solely with the CONTRACTOR.

Drawings must be delivered by AutoCAD standard. Documents must be delivered by MS Office standard. All documents and drawings must contain numbering from both the CONTRACTOR and the OWNER, following the numbering rules to be provided at the start of the project. Document sizes must comply with DIN standards, with the smallest size being A4 (210 x 297 mm) and the largest A0 (841 x 1189 mm).

The BIDDER must consider that they must follow the communication standards, file exchange formats, reports, etc., defined for the project by the OWNER and its management company. Therefore, various reports must be submitted in the formats and standards defined for the project. Examples include:

- Daily Work Reports (RDOs);
- Weekly reports;
- Monthly reports;
- Schedules;
- S Curve;
- Work Breakdown Structures (WBS);
- Activity report
- Monitoring Report
- Work Breakdown Structures (WBS);
- Physical progress curves;
- “As Built” of all issued drawings, etc.

At a minimum, the following documents must be considered as deliverables, included in the project documentation and must be included in the project's data book:

- **Drawings and Document List:** A comprehensive list of all documents and drawings issued for the CCTV system project, indicating the respective revision statuses and dates, in compliance with the numbering and coding standards defined by the OWNER and CONTRACTOR.
- **System Racks and Internal Equipment Distribution:** A detailed list of network connection points per building/cabinet must be provided, indicating the allocation of the new cameras. For each rack, a table specifying the cable lengths (both internal and external), identification tags, and cable types must be

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supplied, along with rack elevation drawings (showing switches, network video recorders – NVRs, patch panels, power supplies, etc.).

- **Photographic record of external optical cabling routes:** A photographic documentation must be prepared covering all camera installation points, external cabling routes, and entry points for each building. The documentation must include route descriptions, and the identification of each segment related to the CCTV infrastructure.
- **Detailed Drawings of Camera Installation Points:** Drawings must be prepared for each building, showing the precise locations of all installed cameras, including point identifiers, cabling routes, mounting details, and associated equipment, such as junction boxes or protective housings.
- **Certification of Cabling and Camera Connections:** All cabling associated with the new camera system, whether metallic or optical, must be evaluated and certified using calibrated and up-to-date equipment. Certification records must follow the cable labeling and identification standards pre-approved by the OWNER, ensuring that all installations meet applicable technical and safety standards.
- **Cabling and System Test Plan:** A detailed test plan must be developed and submitted, outlining the procedures for verifying the integrity and operational readiness of all installed cabling and camera connections, including functional tests of each camera's network communication, power delivery (e.g., PoE), and video streaming capabilities.
- **System Test Plan:** A detailed test plan must be developed and submitted, outlining the procedures for verifying the integrity and operational readiness of all installed or configured implementations, including functional tests of each camera's network communication and video streaming capabilities.

The BIDDER must confirm that, after the preparation of the designs, they will be sent for comments and approval by the OWNER/MANAGER, and after the CONTRACTOR addresses the comments, the designs must be issued to the OWNER as a certified copy. The final certification must be provided after the completion of all tests.

13. TRAINING, COMMISSIONING, AND STARTUP

Training, commissioning, and startup must be included as part of the CONTRACTOR's supply scope for Security CCTVs camera system.

The BIDDER must include training sessions for system operators and administrators covering system operation, camera management, alarm handling, reporting, and maintenance procedures as well as any other system functionalities included in this scope.

The BIDDER must consider, for each implemented system, offering training for three (3) different classes with up to nine (9) people per class and a total of 25 people.

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14. EXPEDITING, INSPECTION, AND TESTING

14.1 EXPEDITING

The OWNER and/or its representative may conduct monthly visits, if deemed necessary, to monitor the project phase, equipment procurement phase, and equipment integration phase. The CONTRACTOR will be required to submit monthly reports detailing project progress and activities conducted each month.

14.2 INSPECTION

The supply scope is subject to inspection by the OWNER for the following items, if deemed necessary:

- Verification of purchased equipment versus the specifications of what was ordered and/or any modifications agreed upon during project development;
- Verification of the use of software licenses and associated media;
- Assembly of equipment in accordance with project documentation;
- Verification of packaging for transportation of equipment to the site.
- Verification of the system configuration according to the OWNER's requirements;
- Verification of the use of software licenses and associated media;

Any implementations or equipment parts that show irreparable damage or defects, or that do not comply with good engineering practices, may be rejected. Acceptance by the OWNER and/or its representative does not eliminate the possibility of subsequent rejection in the event of deficiencies detected after delivery of the equipment.

14.3 ACCEPTANCE TESTS

The acceptance tests will verify the correct operation of the system with all software components, integrations, and configurations fully implemented. A period of two (2) weeks must be allocated for testing, including all related expenses.

The SUPPLIER must submit, as part of the project documentation, a software acceptance test procedure, which must include at a minimum:

- Verification of equipment assembly according to project documentation;
- Verification of equipment identification;
- System energization and verification of functionality (both visual and through system diagnostics);
- Verification of system configuration, functionalities, alarms, events, etc.
- Verification of software installation and configuration according to the approved project documentation;
- Verification of software component identification, including version control and license validation;

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- System initialization and functional validation, performed both through the user interface and through system diagnostic tools;
- Verification of system configuration parameters, analytics, workflows, permissions, alarms, events, integrations, and all functional modules.

The CONTRACTOR's testing procedures must be submitted in advance for the OWNER's approval.

15. THIRD-PARTY SYSTEMS INTEGRATION AND TESTING

The BIDDER must also consider the connection, configuration and validation of integrations with external systems such as Access Control System for event association, or client-provided applications (Access Control System – Schneider Ecostruxure Security Expert). The Access Control integration must include the relevant licensing for receiving and reacting to triggers from the Access Control System (EcoStruxure Security Expert) for specific monitoring conditions. The number of cameras integrated into the system shall be the same as the total number of doors monitored by the access control system.

The BIDDER must also consider monitoring the integration and communication tests between the fire protection systems and the OWNER's corporate system.

16. STARTUP, ASSISTED OPERATION, AND FINAL ACCEPTANCE OF TESTS

The Startup will mark the beginning of the System's operation in each area during the various phases of installation.

The assisted operation corresponds to a period of one (1) month for evaluating the System's performance and addressing any issues detected by users.

The final system acceptance will be the final phase and will occur when all equipment is installed, with all functionalities implemented and fully operational.

Comprehensive reports detailing all functionalities and statistics must be provided, encompassing all installations required for the final system acceptance.

17. WARRANTY

The CONTRACTOR must guarantee that the supplied equipment and services meet all the requirements of this technical specification and its application.

The CONTRACTOR must provide a warranty of at least thirty-six (36) months after installation of equipment and configuration services. Any costs associated with services during the warranty period will be the responsibility of the CONTRACTOR.

The BIDDER must present an SLA (Service Level Agreement) for warranty support for the OWNER's analysis.

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18. INTERFACES

The BIDDER must consider as part of its scope the management of all interfaces, including with THIRD PARTIES involved, whose coordination will be carried out by the MANAGER and/or OWNER, also providing and exchanging information and data for the correct planning of execution times and necessary data for the development of the Project and the proper installation, operation, and delivery of the contracted scope.

19. PROJECT MANAGEMENT AND CONTROL

To support project follow-up meetings, the SUPPLIER must present monthly reports detailing the main activities carried out during the period, as well as all updated planning documentation, containing at least the following elements:

- Progress Map;
- Documents produced during the period;
- Progress Curves ("S" curve);
- Update Procurement Map;
- Production quantities;
- Main issues encountered and resolution strategy for the next period;
- Comparison between planned vs. actual;
- Critical path;
- Pending items;
- Key activities from the previous period.

The SUPPLIER must prepare a detailed schedule of its activities in an electronic file using MS Project software.

The schedule must indicate the main activities related to notable events such as:

- Issuance of engineering documents;
- Equipment procurement;
- Release for shipment;
- Equipment delivery;
- Mobilization;
- Assembly status;
- Commissioning;
- Startup;
- Assisted operation.

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20. SUPPLY BY THE OWNER

Any supply items necessary for the complete and proper implementation of the project, and not explicitly listed below in this item (Owner's Scope) as the OWNER's responsibility, will be the responsibility of the CONTRACTOR, including but not limited to materials, equipment, tools, and services.

Thus, the following items will be the exclusive responsibility of the OWNER for this contract:

- Space for the installation of the CONTRACTOR's site facilities, duly licensed;
- Provision of water and electricity access points near the site facilities, for their supply, with the CONTRACTOR responsible for voltage step-down, interconnection, and internal distribution within those areas;
- Provision, at the utility interconnection points that are to be serviced by the CONTRACTOR (electricity, water, drainage, sewage, etc.), as necessary for contract fulfillment.

21. CONTRACTOR'S OBLIGATIONS

The BIDDER must consider the following as part of the CONTRACTOR's Responsibilities and Obligations:

- Full compliance with the OWNER's standards and procedures;
- Execution of contracted services according to the physical schedule that is part of the contract;
- Responsibility for losses and damage caused to people or property of the OWNER or third parties, directly or indirectly resulting from the execution of the contracted services. To this end, the CONTRACTOR must maintain civil liability insurance consistent with the value of the project. The premium for this insurance will be fully borne by the BIDDER, who must indicate in its proposal all values considered in the future policy, such as: insured amounts, third-party insurance, premiums, and other relevant but not mentioned items;
- Provide the OWNER with reports about the work/service in the requested format, containing production, appropriation, and other technical data related to the progress of the services, including administrative information;
- Submit, considering the INSS registration number provided by the OWNER, when requested, the following:
 - Declaration and Information on Construction – DISO, in two (2) copies;
 - Construction Registration Certificate (CMA) – copy ;
 - CGC/CNPJ card (copy);
 - Technical Responsibility Annotation (ART) from CREA (copy);
 - All GRPS/GPS related to the construction (own and subcontractors') – copies, listed on the back of the DISO or on a separate sheet, one for each contractor, in two (2) copies;

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- All Construction Invoices/Bills;
- Payment of social security obligations directly under the CEI opened by the OWNER, specific to the project.
- INSS Clearance Certificate:
 - The BIDDER must consider that after contracting, the OWNER will open a registration number for the CONTRACTOR, who must present the appropriate proof of registration and payments to the INSS;
 - Thus, companies must indicate in their contracted services, in addition to their name, the description of the areas to be constructed, areas to be demolished, areas for external activities, among others, with the corresponding INSS registration stamp and control.

22. FINAL CONSIDERATIONS

This specification and its attached documents were prepared to assist in project preparation, service quotation, equipment, and materials by qualified technicians of the BIDDER and are not intended to be exhaustive or include all details. The BIDDER must familiarize itself with the information described in this specification and accept full responsibility for the project as the CONTRACTOR. The BIDDER must add or complete in its proposal any information not mentioned, missing, incomplete, or incorrect in this specification.

The quantities and specifications presented in this document are estimates based on currently available information. The actual and final quantities will be determined during the project detailing phase. Additions, reductions, or changes, if necessary, will be negotiated based on agreed unit prices to allow value adjustments. A unit price list for all system components must be part of the final contract.

The BIDDER must review all content in this specification, including its annexes, to verify the accuracy of the information for the proper execution of the project. If the BIDDER notices discrepancies, errors, omissions, or any other conditions that may affect the nature of the project and its execution, or the future performance of the system to be implemented, the BIDDER must consult the OWNER to clarify any doubts within the pre-established technical query period.

Failure to identify and formally communicate the above-mentioned items to the OWNER within the predetermined period is solely and exclusively at the BIDDER's risk.